

CSE370

L-1

(Self notes)

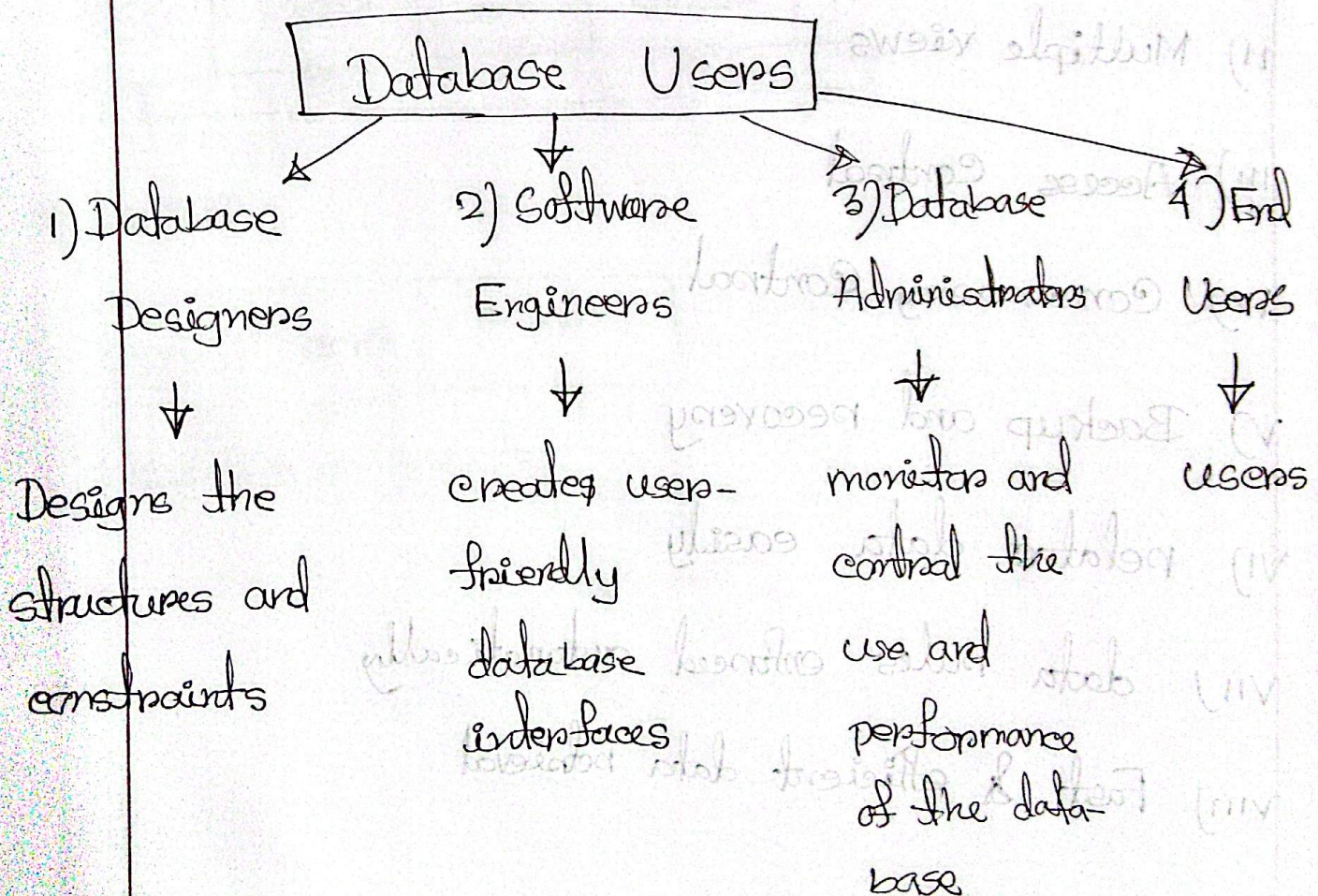
* Database → organized collection of related data

Advantages :

- i) Less data redundancy - No duplicates
- ii) Multiple views
- iii) Access Control
- iv) Concurrency Control
- v) Backup and recovery
- vi) relates data easily
- vii) data rules enforced automatically
- viii) Fast & efficient data retrieval

Disadvantages :

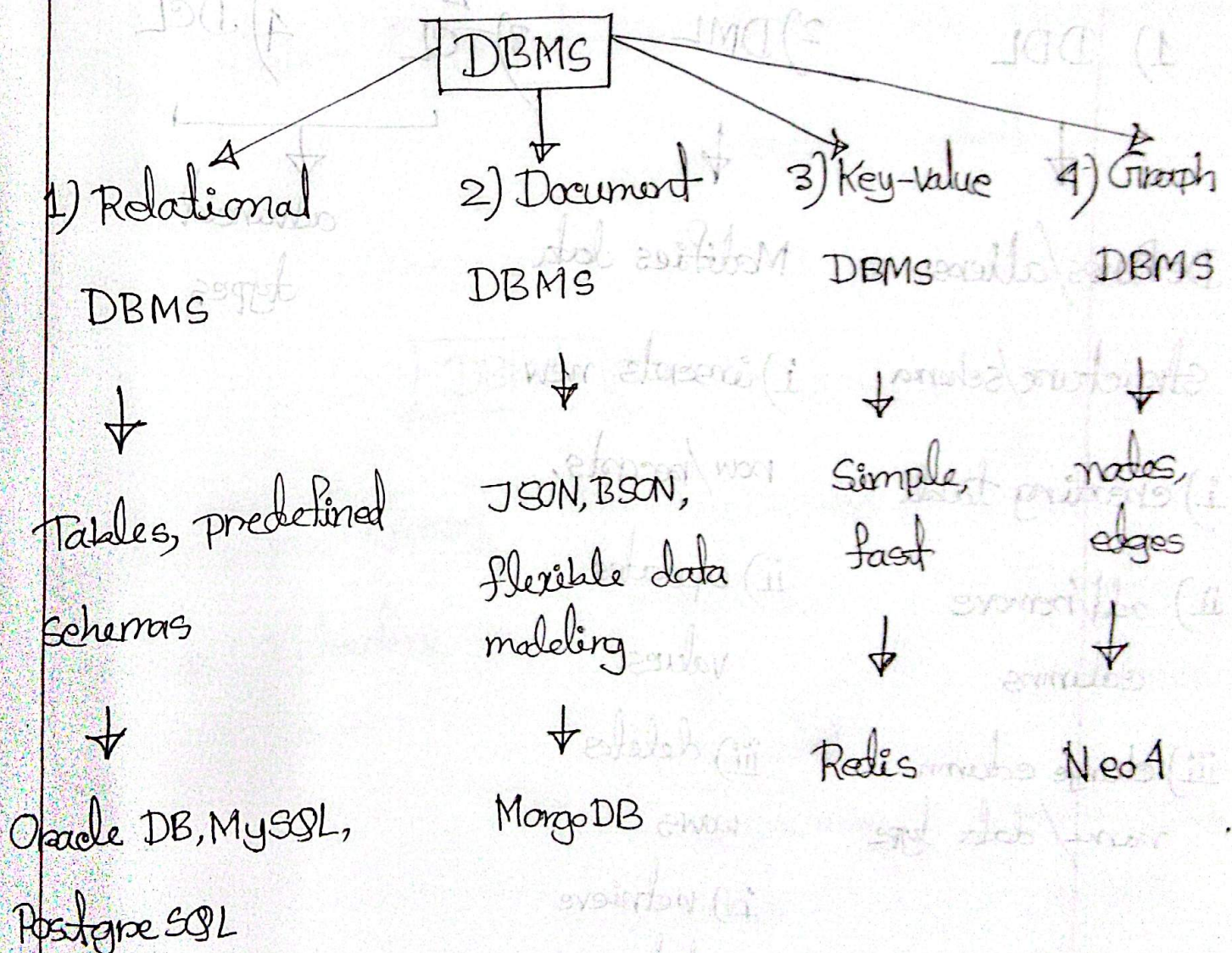
- i) High initial cost
- ii) Extra hardware
- iii) Overhead
- iv) Maintenance cost



* **DBMS** → Database Management Systems



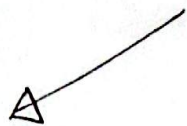
A software package to help create and maintain the databases



DBMS Languages



SQL (Structured Query Language)



1) DDL



Defines/alters

structure/schema

i) creating table

ii) add/remove columns

iii) change column name/data type



2) DML



Modifies data

i) inserts new

row/records,

ii) updates values

iii) deletes

rows

iv) retrieve data



3) TCL



advanced types



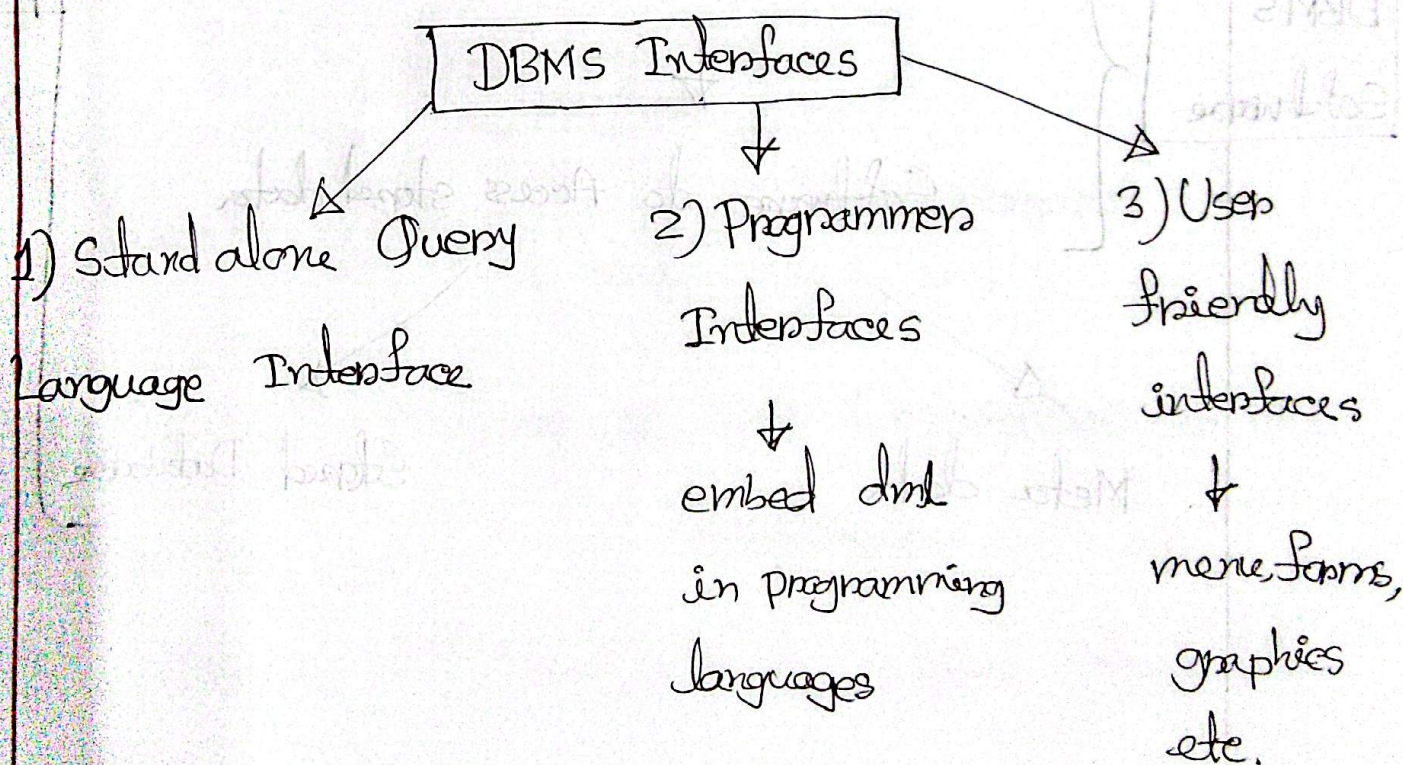
4) DCL

* DDL (Data Definition Language)

SQL command → Create Table, Alter Table

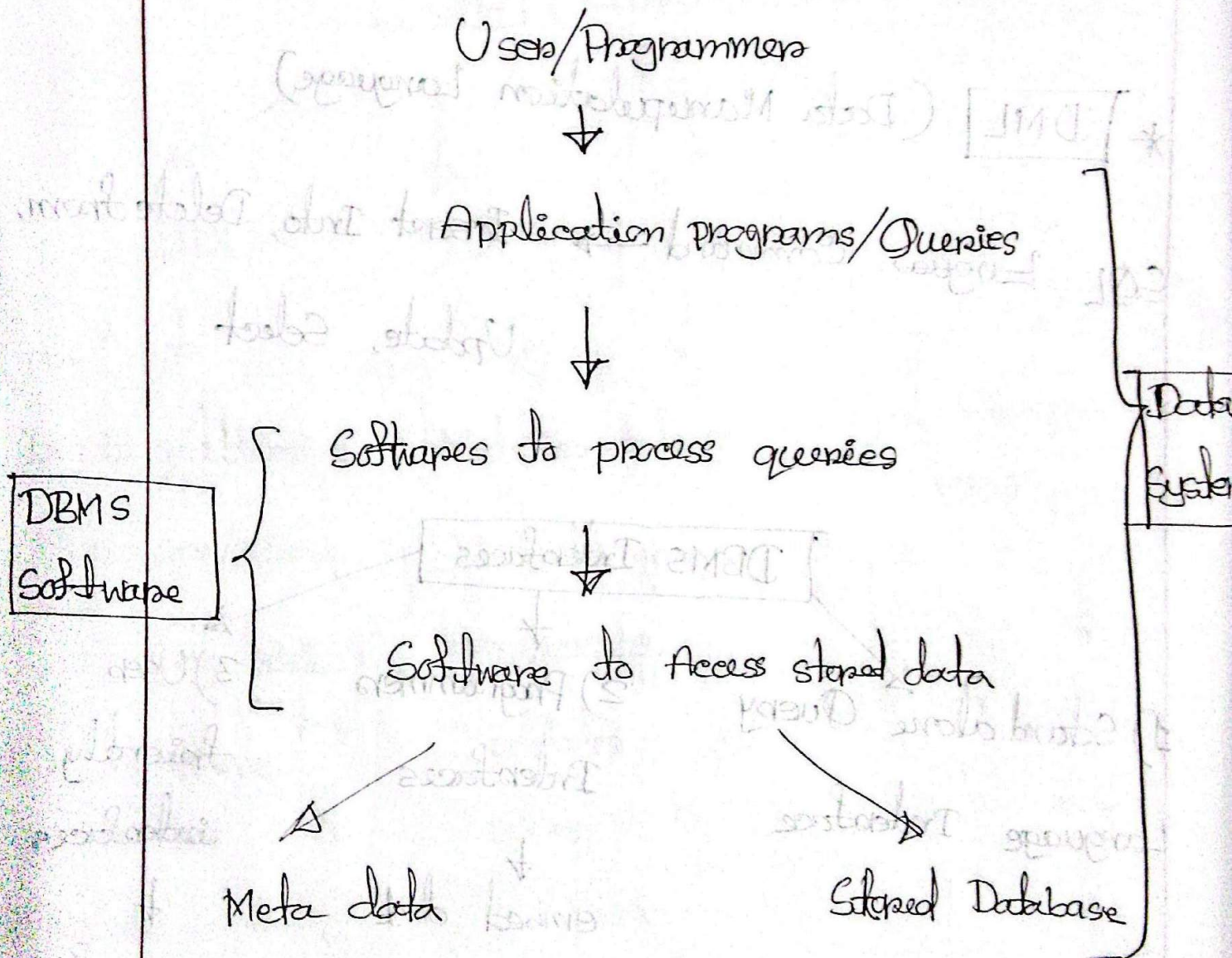
* DML (Data Manipulation Language)

SQL Language Command → Insert Into, Delete From,
Update, Select



Database System → Software + data

* Database System Environment :



Full stack for a Database

web application:

UI/ Frontend

→ HTML/CSS, React

Backend

→ PHP, Django, Node.js

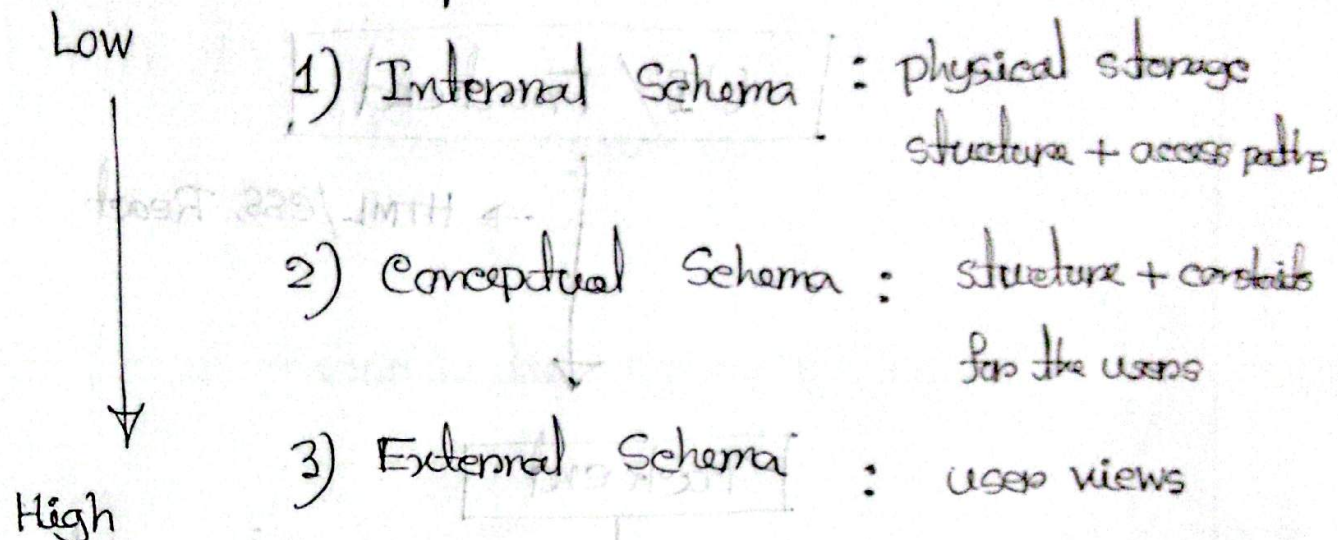
Database

→ MySQL, MongoDB

Database [Schema] → Desp Description of the database

Database [state/instance] → stored actual data

Three - Schema Architecture



Data Independence

1) Logical Data Independence

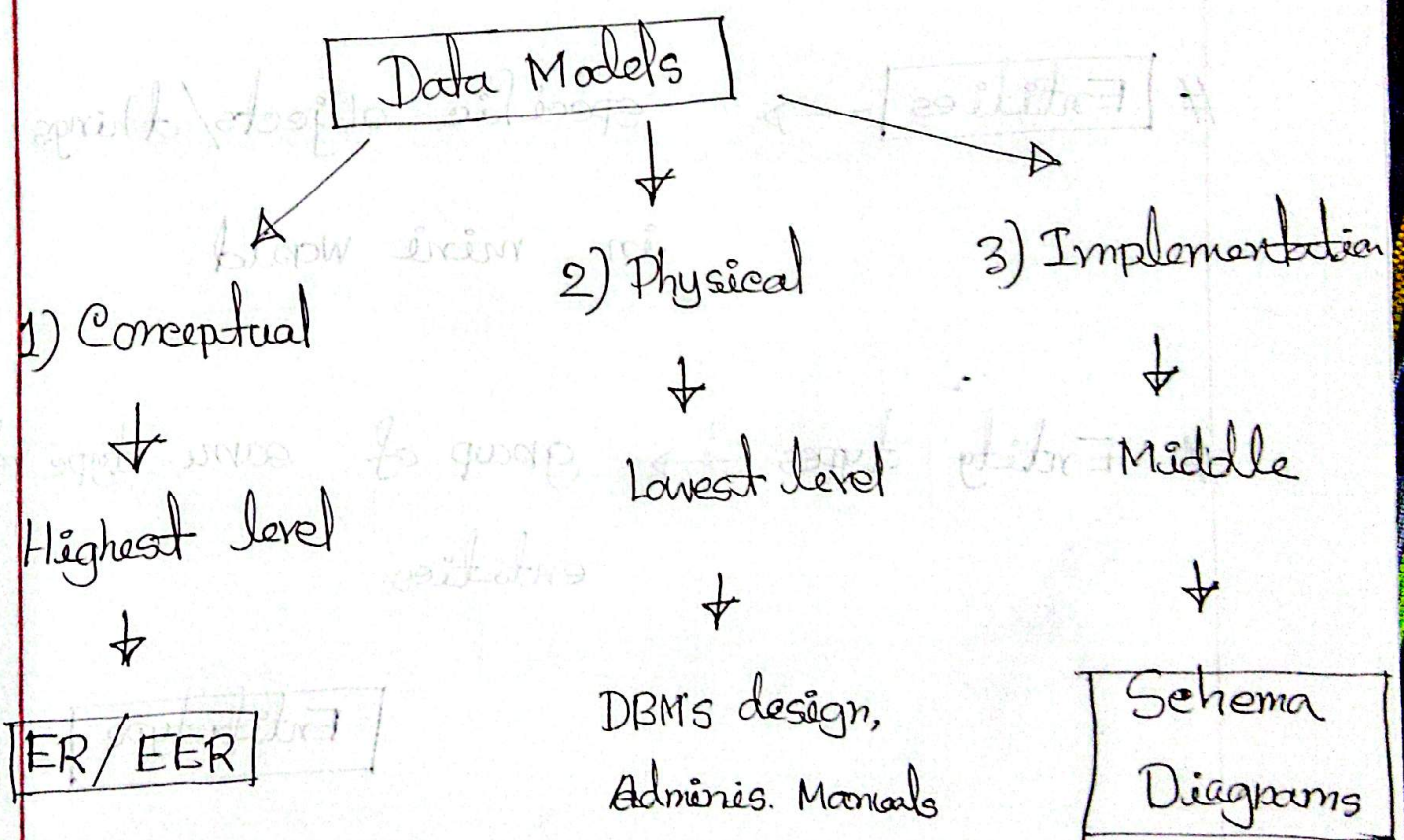
↳ change at conceptual level, no impact at the external level

2) Physical Data Independence

↳ change at the physical internal level, no impact at the conceptual level

* **Data Abstraction** → hiding storage details and highlighting key essentials.

→ Achieved through Data Models



Lecture-2

ER (Entity Relationship)

Mini World → a system w/ from real world scenario

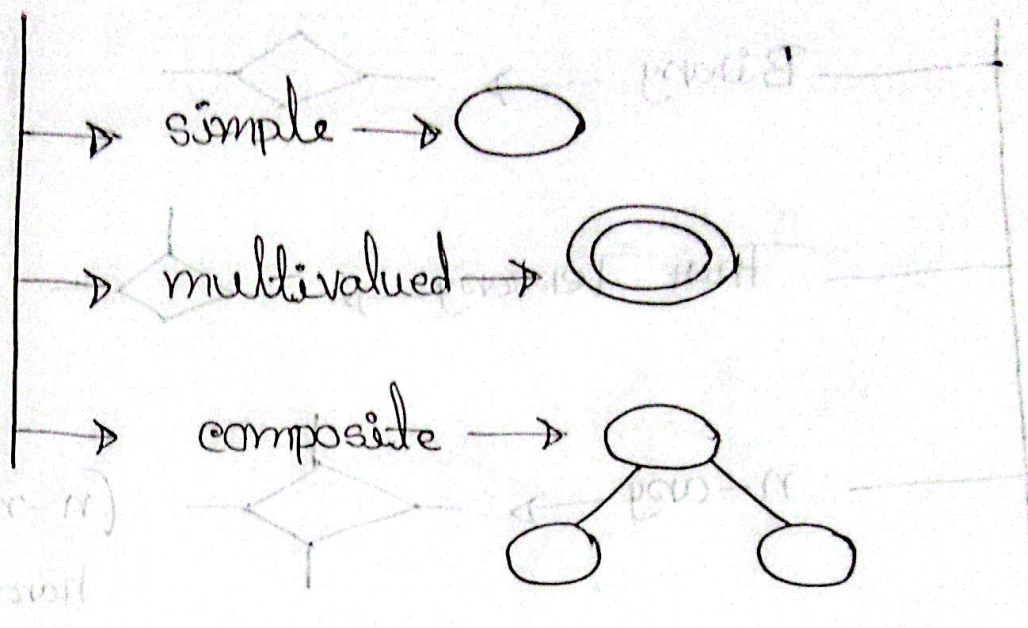
Ex → Train reservation system

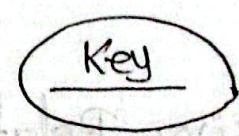
Entities → specific objects/things in mini world

Entity types → group of same type of entities

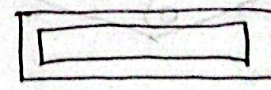
Entity type

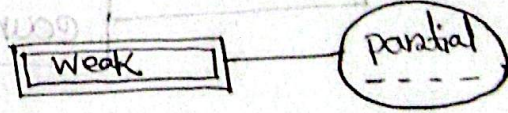
Attributes → 

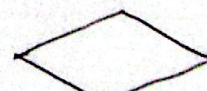


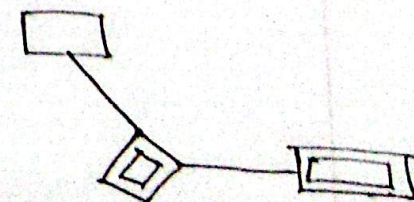
Key Attribute → 

Derived Attribute → 

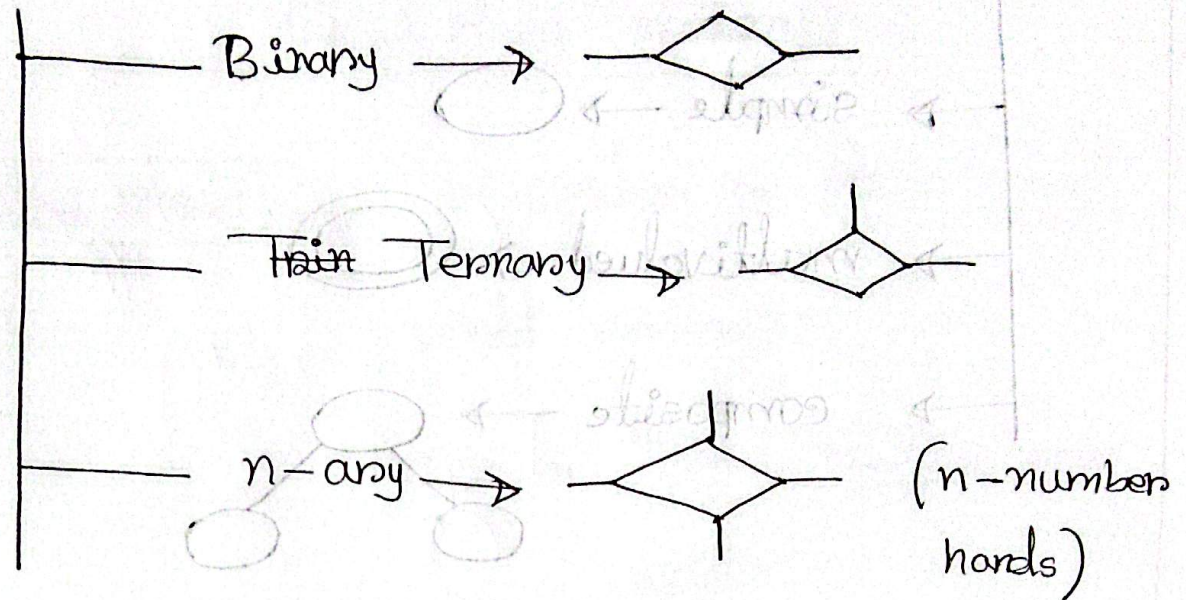
Weak Entity → 

Partial Key → 

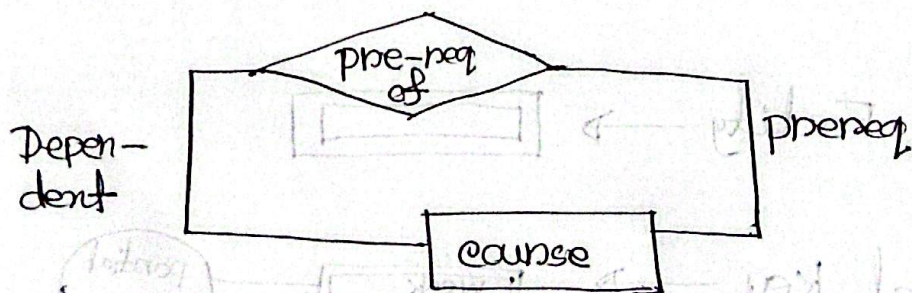
Relationships → 

Identifying Relationships → 

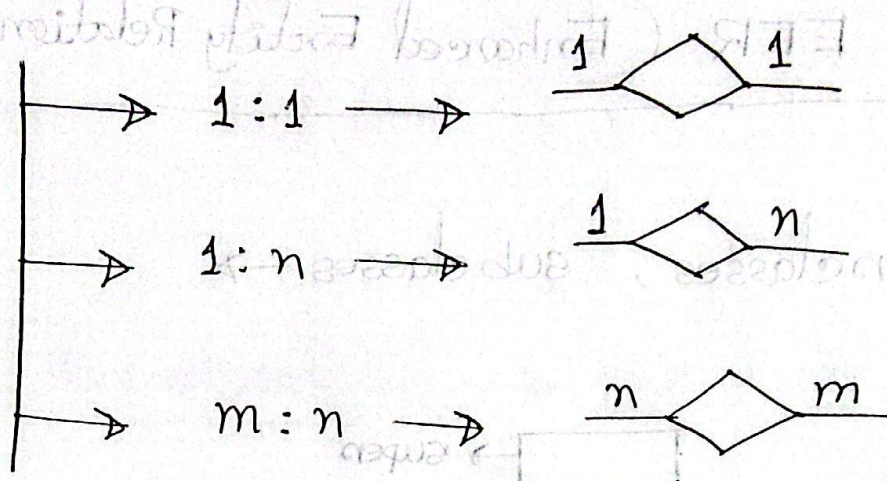
Degree of relationships



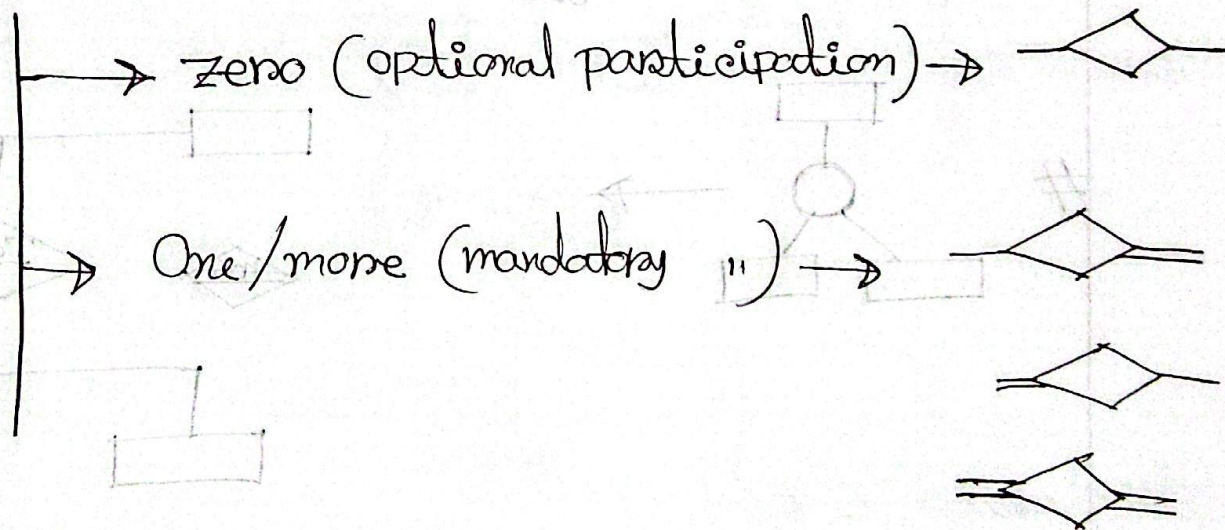
Recursive Relationship



Cardinality Ratio



participation constraints

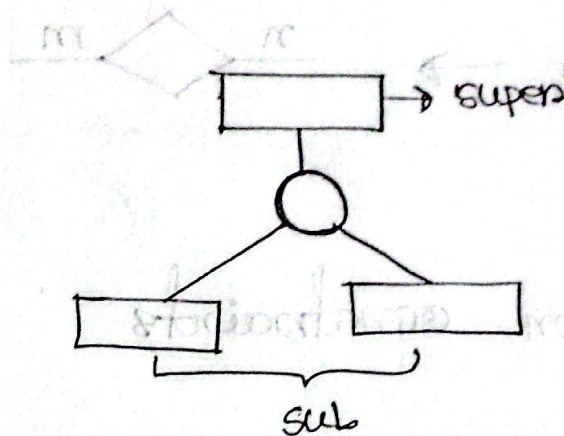


TE-A relationship

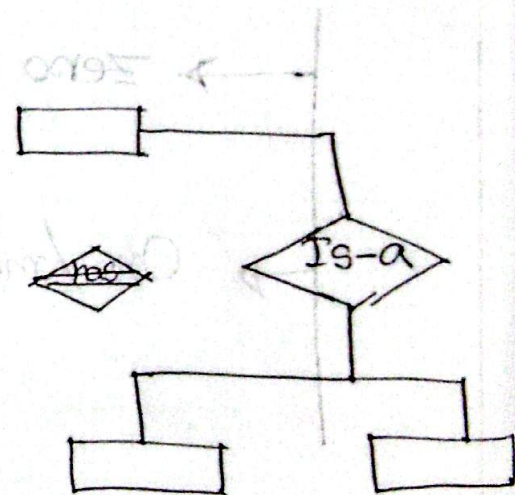
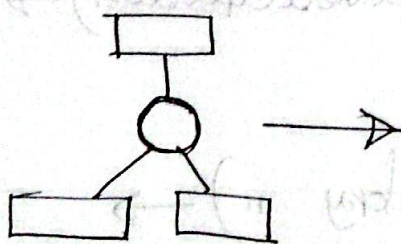
Lecture-3

EFER (Enhanced Entity Relationship)

Superclasses, subclasses →



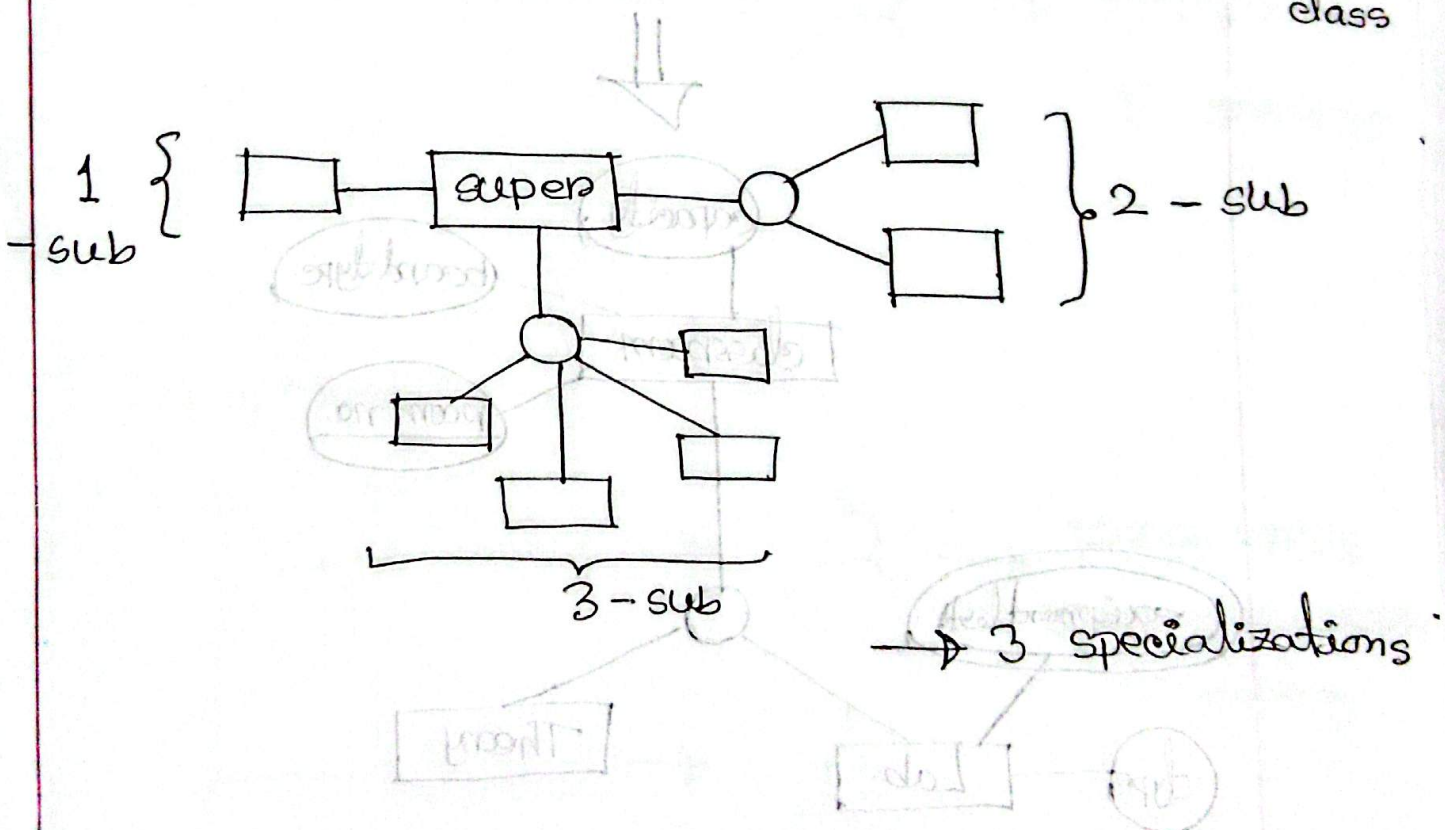
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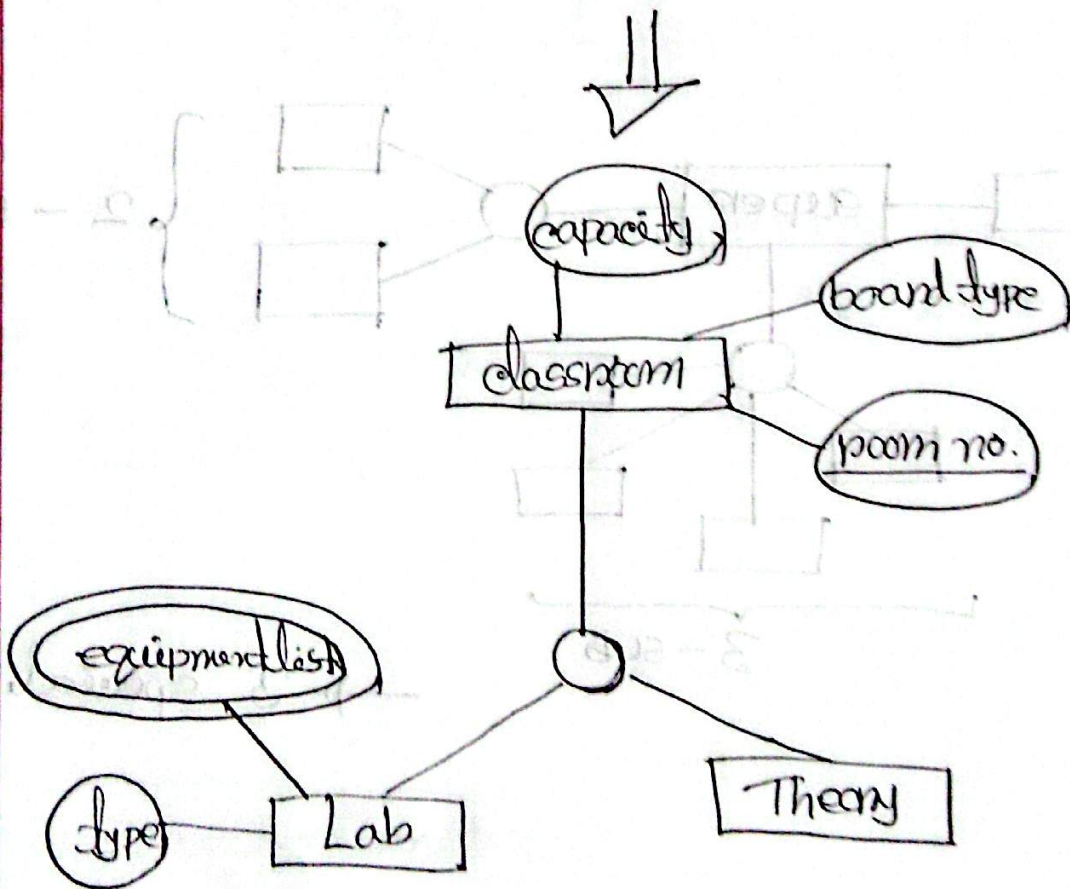
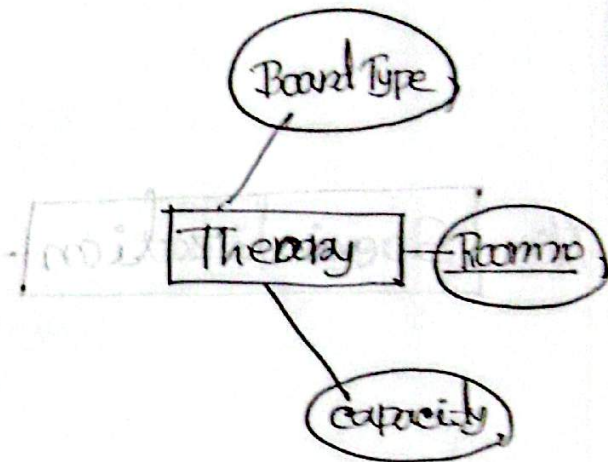
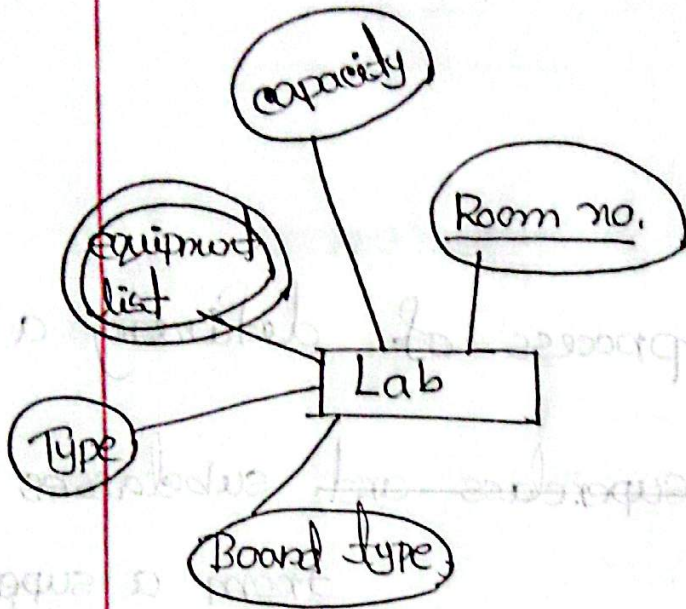
IS-A relationship

A subclass inherits all attributes of its superclass

Specialization → process of defining a set of superclass and subclasses from a super class

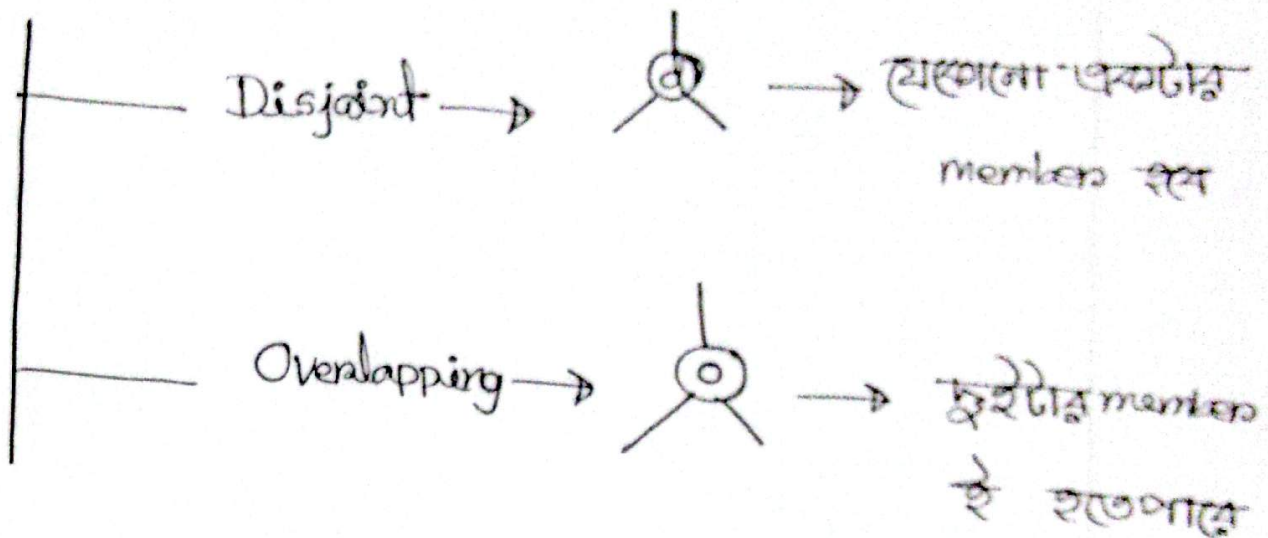


Generalization → process opposite of specialization

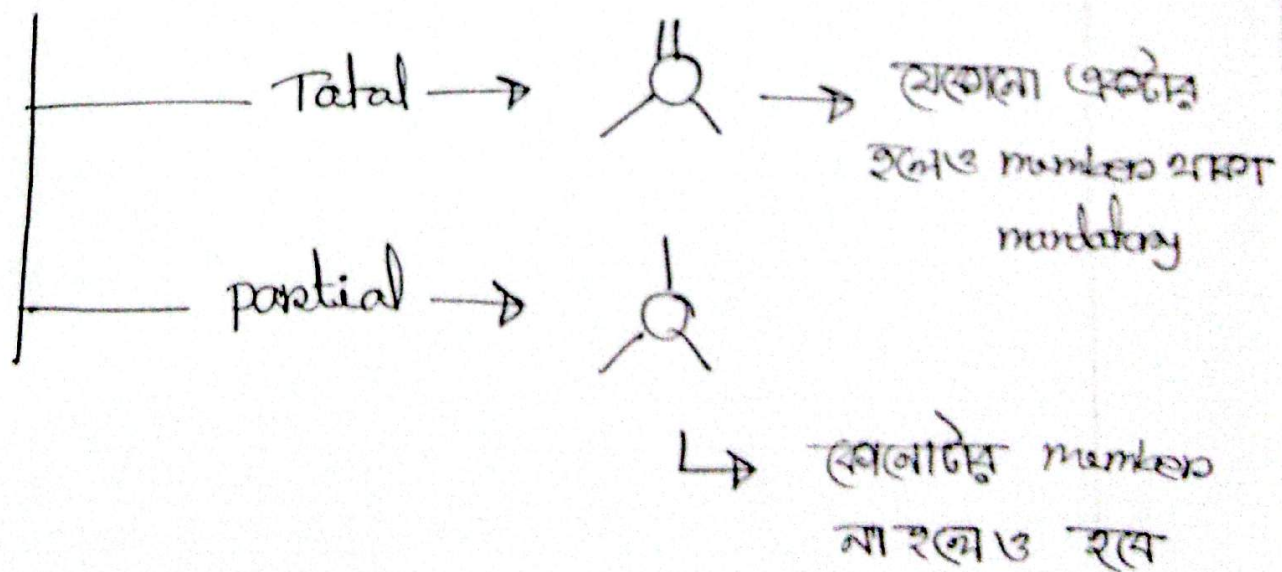


Constraints on Specialization/Generalization

1) Disjointness :



2) Completeness :



Lecture-4

Relational Data Model & Database Constraints

Relation $\rightarrow$ Table

~~Co~~ key $\rightarrow$ $\cap$ Attributes $\rightarrow$ column headers

Tuples $\rightarrow$ rows

Rows
~~Tuples~~, are not ordered but attributes are

Relation Schema $\rightarrow$ description of the table



$R(A_1, A_2, A_3, \dots, A_n)$



Example: Student (Name, ID, Email)

Domain → set of all eligible values of an attribute

→ includes data type

→ may include formats and ranges

Key Concepts

Super key → set of attributes whose ~~that~~ are unique to each tuple

Key → minimal superkey

Candidate keys → key All keys

Primary Key → One unique key to candidate key to identify the table

Foreign key → Reference to the primary key of another table

→ Foreign key ~~ये~~ table १

→ **referencing relation**

→ primary key " " " **referenced relation**

Relations in an ER/EER diagram and in a relational data model is a not are not same.

Schema vs. State :

Student

<u>SID</u>	Name	Email
01	Abir	abir@gmail
02	Ananya	anany@gmail

Student

<u>SID</u>	Name	Email
01	Abir	abir@gmail
02	Ananya	anany@gmail

Fig1 : Schema

Fig2 : State

#

Relational Integrity

Constraints

1) Domain
constraint

2) Key
constraint

3) Entity
Integrity
constraint

4) Referential
Integrity
constraint

Domain key constraint → Attribute - এর

data type হাড়া অন্য কোনো type এর data insert করতে করা যাবে না।

Key constraint → primary key এর

কোনো value duplicate হবেনা।

Entity Integrity Constraint → primary

key - এর কোনো value 'Null' হবেনা।

Referential Integrity Constraint →

1) ~~referenced~~ foreign key column - এর কোনো value referenced column এর relation - এর primary key column - এর বাইরে হবেনা।

2) Foreign key যদি referencing table relation
এবং primary key হয়, তাহলে null allowed
হবেনা।

Enforcing Relational Integrity Constraints:

1) While Inserting these constraints may
be violated $\rightarrow$

i) Domain constraint

Solⁿ: Declaring data type

ii) key constraint

Solⁿ: Declaring primary key and
using the unique keyword

iii) Entity Integrity Cons.

Solⁿ: Declaring primary key

iv) Referential Integrity Cons:

Solⁿ: Declaring Foreign keys
with proper references

2) While Deleting these constraints may
be violated $\Rightarrow$

i) Referential Integrity Cons. :

Solⁿ: referencing table এ আছে

এক value referenced table

এক delete করা দিলে হয়।

i) Foreign key declaration

ii) On delete $\rightarrow$ Restrict
 $\rightarrow$ cascade
 $\rightarrow$ set null

~~iii)~~ 3) While Updating these constraints may be violated →

i) (same as 'Insertion')

ii) "

iii) "

iv) Referential Integrity Constraint

Soln: Referencing and Referenced both relation update ~~are~~
जैसे this constraint may be violated.

i) Foreign key declaration

ii) On update
→ restrict
→ cascade
→ set null